

Age of Information for Periodic Status Updates Under Sequence Based Scheduling

Fang Liu¹, Wing Shing Wong², *Life Fellow, IEEE*, Yuan-Hsun Lo³, *Member, IEEE*,
Yijin Zhang⁴, *Senior Member, IEEE*, Chung Shue Chen⁵, *Senior Member, IEEE*,
and Guoliang Xing⁶, *Fellow, IEEE*

Abstract—This paper considers a system in which multiple users send periodically generated status information to a common access point (AP) over a collision channel. To avoid high overhead, there is no time synchronization and no feedback information from the AP to indicate whether a transmission is successful or not. The performance metric that we focus on is the age-of-information (AoI), which represents the freshness of the status information received at the AP. For this model, we propose a sequence based MAC scheme in which each user is pre-assigned a periodic sequence to schedule transmissions. This scheme guarantees each user at least one successful packet transmission within a sequence period, in the absence of time synchronization and feedback information from the AP. To the best of our knowledge, this is the first study investigating AoI performance under a sequence based MAC scheme. We derive the closed-form expressions for average AoI, average peak AoI and average age penalty under the sequence based scheduling. Besides, we derive several critical properties of the sequences to optimize the AoI performance. Comparison results show that our proposed sequence scheme outperforms slotted ALOHA and framed ALOHA in various settings.

Index Terms—Age of information, sequence based scheduling scheme, periodic status updates, random access.

I. INTRODUCTION

IN RECENT years, with the rapid development of Internet of Things (IoT) and the 5th Generation Mobile Communication (5G) technologies, many new applications have

emerged and expect real-time information updates. Conventional performance metrics, such as throughput and delay, cannot accurately reflect the information freshness in these real-time applications. To this end, a metric called the *age-of-information* (AoI), was proposed by [1] to indicate the freshness of the status updates. Specifically, AoI measures the time elapsed since the generation of the last successfully received status update.

In this paper, we focus on AoI performance in the scenario where multiple users transmit periodically generated packets to an access point (AP) without feedback. This model is commonly found in practical applications. A typical scenario of periodic traffic is periodic wireless sensor networks (PWSNs) [2], in which wireless sensors collect status updates from a specific region in a periodic manner and then send the updates periodically to AP for environment surveillance. Periodic traffic can also be found in industrial control systems, traffic monitoring systems, and so on. Transmission without feedback is desirable for low-power devices with limited resources, such as sensor networks, as it reduces the energy consumption and resource usage of the network. Additionally, transmission without feedback is also suitable for long-distance transmission to reduce delay and bandwidth occupancy, such as ground-satellite communications [3] and WiFi-based long distance mesh networks [4]. Moreover, there is no time synchronization in the system we consider. It is challenging to achieve accurate synchronization since it usually imposes demands on hardware complexity and requires frequent exchanges of messages, which would lead to heavy overhead [5]. Furthermore, the system does not require carrier sensing and successive interference cancellation (SIC) techniques or similar that may incur extra overheads. It should be noted that most of the existing works are not applicable to our model.

For this system, devising proper multi-access schemes is crucial for achieving optimal information freshness. In this paper, we focus on deterministic schemes, which can be regarded as sequence schemes in which each user is pre-assigned a transmission schedule in the form of a binary sequence. At each time slot, every user reads out its current sequence value and then conducts an action (either transmitting through the channel or keeping silent) accordingly. The main reasons for our interests in the sequence based schemes are as follows. First, deterministic schemes do not depend on channel monitoring and feedback information from the AP about the

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Fang Liu, Wing Shing Wong, and Guoliang Xing are with the Department of Information Engineering, The Chinese University of Hong Kong (CUHK), Hong Kong, SAR (e-mail: fangliu@cuhk.edu.hk; wswong@ie.cuhk.edu.hk; glxing@cuhk.edu.hk).

Yuan-Hsun Lo is with the Department of Applied Mathematics, National Pingtung University, Pingtung 900391, Taiwan (e-mail: yhlo@mail.nptu.edu.tw).

Yijin Zhang is with the School of Electronic and Optical Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: yijin.zhang@gmail.com).

Chung Shue Chen is with the Nokia Bell Laboratories, Paris-Saclay Center, 91300 Massy, France (e-mail: chung_shue.chen@nokia-bell-labs.com).

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result of each transmission. Such simplicity in implementation is particularly desirable in our investigated system. Second, previous studies [6], [7], [8], [9], [10] have offered extensive comparison results on delay and throughput for sequence based schemes and other existing schemes. For example, the sequence based schemes can achieve shorter delay while maintaining the same average throughput, compared with the optimal slotted ALOHA scheme. Furthermore, an appropriately designed schedule sequence set can guarantee at least one collision-free transmission within a sequence period. On the contrary, most of the existing methods fail to provide such a hard guarantee owing to their probabilistic nature. Nevertheless, the AoI performance under sequence based schemes as well as the comparisons with existing schemes have not been studied yet. To the best of our knowledge, this is the first paper that investigates AoI performance of sequence based multi-access schemes. For our investigated model, we will consider two commonly used schemes as the baseline references: one is the slotted ALOHA scheme studied in [11], and the other one is the framed ALOHA scheme [12]. Under the slotted ALOHA scheme, each user transmits to the AP at each time slot with a certain probability. Under the framed ALOHA scheme, at the beginning of each status generation period, each user randomly picks a certain number of slots within a period for packet transmission. The two ALOHA-based schemes are well-studied in the literature and easy for implementation, and thus often serve as benchmarks for evaluating the performance of newly proposed schemes.

The first and foremost goal of this paper is to analyze AoI performance of the sequence based scheme. We will also show the superiority of the sequence based scheme over the slotted ALOHA and the framed ALOHA schemes in comparisons. It is expected that the results reported in this paper can inspire new designs and motivate further research interests on enhancing the sequence based schemes. The main contributions of this paper are summarized below.

- 1) We propose a novel sequence based scheduling scheme for periodic status updates over a collision channel without feedback. To the best of our knowledge, this is the first paper that looks into the design of sequence based scheduling and carries out an analytical study in depth, in order to enhance the AoI performance over today's conventional schemes such as slotted ALOHA and framed ALOHA. For the proposed sequence based scheduling scheme, we derive the closed-form expressions for average AoI, average peak AoI and average AoI penalty.
- 2) We find several interesting structural properties of the sequence based scheme, which are critical for optimizing the AoI performance of systems with periodic status updates. In the case where one constructs the schedule sequences with period same as the status generation period, we find that it is important to allocate a transmission slot to each user at the beginning of each scheduling period. With this property, the search efficiency for the optimal sequences can be greatly enhanced. We also extend our discussions to the case where the period of the schedule sequences is different from the status

generation period. For this case, we find a periodicity property of the sequences that one can use for the optimal design of sequence based scheme.

- 3) We conduct a comprehensive comparative study among slotted ALOHA, framed ALOHA, and our proposed sequence based scheme, under various system settings including different levels of time synchronization. We focus on their key performance metrics including the average AoI, average peak AoI, and average AoI penalty and conduct extensive computer simulations in parallel to evaluate. Results show that in comparison to slotted ALOHA and framed ALOHA, the proposed sequence based scheme achieves superior AoI performance and also leads to higher energy efficiency.

The rest of this paper is organized as follows. Section II introduces related works of AoI and sequence based schemes. Section III describes the system model. In Section IV, we present some preliminaries and necessary technical definitions of sequence based scheme. The AoI performance under sequence based scheme is analyzed in Section V. In Section VI, we focus on optimizing the AoI performance under the sequence scheme. An extended case for the sequence scheme is discussed in Section VII. The comparison results among the sequence scheme and two baseline schemes are shown in Section VIII. Finally, we conclude the paper in Section IX.

II. RELATED WORKS

AoI has gained tremendous research interests in recent years. It was first introduced in [1] for vehicular communications as a metric to measure the information freshness at the destination side. Subsequently, research works on AoI emerged in plenty of application areas. In the following, we will concentrate on the review of AoI works in multi-access scenarios.

Periodic packet arrival is commonly found in the literature [2], [11], [13]. For example, [11] investigates AoI performance of periodically generated packets under slotted ALOHA, where each user keeps using a fixed transmission probability in the absence of feedback information from the AP. For periodic traffic in the systems where feedback information is available to the users, AoI performance under slotted ALOHA has also been studied. For instance, the information of the number of contending users is assumed to be provided by the AP in [2], so that each user can adjust its transmission probability accordingly. Apart from periodic traffic, generate-at-will pattern [14] and traffic pattern following various distributions [15] such as Poisson arrival have also been thoroughly considered in a large number of works.

Various schemes have been proposed to enhance AoI performance in the multi-access scenarios, especially uncoordinated scheduling schemes in which the users have no need to cooperate with each other. A lot of them are ALOHA-based schemes. For example, the authors in [14] design a threshold-based age-dependent slotted-ALOHA scheme for generate-at-will status generation model, provided that each user is able to know whether its instantaneous age exceeds a predetermined

threshold. Frame Slotted ALOHA with Reservation and Data slots (FSA-RD) is proposed in [16], where a reservation slot consisting of multiple mini-slots is introduced for users to contend for the data slots. There are also many works based on carrier sense multiple access (CSMA), where a user attempts to determine whether another transmission is in progress using a carrier sensing mechanism before initiating its own transmission. Mini-slotted threshold ALOHA (MiSTA) is introduced in [17], which is age-aware and works in a CSMA-like style. In MiSTA, each time slot is preceded by a mini slot during which users with ages exceeding a certain threshold are allowed to test the channel. CSMA-type multi-access schemes with and without preemption for Poisson arrival are investigated in [15]. Moreover, [18] also studies Poisson arrival, and each update has an independent exponential transmission time. Exponential arrival under orthogonal multiple access (OMA) and non-orthogonal multiple access (NOMA) are considered in [19]. Besides the uncoordinated schemes presented above, schemes in which the users cooperate with each other to avoid collisions have also been studied, such as the work in [20].

Since we focus on AoI under sequence based multi-access scheme in this paper, we will also introduce related works for this area in the following. Among existing works on sequence based multi-access, research interests for synchronous sequences are mainly on topology-transparent schedule (TTS) sequences [21] which aim to improve network throughput and transmission delay by utilizing sparsity information. In contrast to synchronous sequences, asynchronous sequences are more challenging and have received more attention in the literature. As an important class of asynchronous sequences, protocol sequences were first proposed in [6] for packet transmissions in a collision channel without feedback. Protocol sequences are binary and periodic, and are designed with favorable Hamming cross-correlation property suitable for multi-user transmission without any coordination. Subsequent works on protocol sequences are mainly on the following two fronts: shift-invariance (SI) and user irrepressibility (UI). SI sequences guarantee that the throughput of each user is invariant for arbitrary starting time offsets among the users. UI sequences guarantee that any user can successfully transmit at least one packet within each predefined tolerable period, for any starting time offsets among the users. Various constructions of SI and UI sequences have been reported [22], [23]. In addition to the multi-access scenario, sequence based schemes have also been widely studied in other scenarios, such as the channel hopping sequences for rendezvous process in cognitive radio networks [24] and the wake-up sequences for neighbor discovery in sensor networks [25].

It is worth noting that existing works on sequence based schemes mainly aim at maximizing the system throughput or minimizing the transmission delay. The work presented in this paper is the first focusing on the AoI performance of sequence based schemes, as far as we know.

III. SYSTEM MODEL

A. System Description

Consider a time-slotted multi-access network consisting of N users and an access point (AP). The time slots are

of equal duration. We denote by $t = 0, 1, 2, \dots$ the indices of the time slots. Every user periodically generates a packet for status update every T time slots, where T is called the *status generation period*. Each status should be delivered to the AP successfully within its generation period, otherwise it will expire and be discarded at the end of the period. The packets transmitted in different time slots have the same size, and the transmission delay of each packet from the user to the AP is bounded by the length of a slot.

We mainly consider a perfect collision channel, in other words, if multiple users transmit their packets at the same slot, then a collision would occur and no packet sent at this time slot can be successfully received. On the other hand, if only one user transmits at a time slot, the transmitted packet can be successfully received by the AP. Besides, after a packet transmission, there is no feedback from the AP to the transmitter to indicate whether this transmission is successful or not.

In this paper, we consider a sequence based multi-access scheme. Each user is pre-assigned a unique binary and periodic sequence to schedule its transmissions. The sequences are denoted by $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$. For simplicity, we assume that these sequences have a common period, which is denoted by L . The properties of the sequences will be introduced in Section IV in detail. We denote the sequence assigned to user i by $s_i = [s_i(0) \ s_i(1) \ \dots \ s_i(L-1)]$, where $s_i(j) \in \{0, 1\}$, $j \in \mathbb{Z}_L$, for $i \in \{1, 2, \dots, N\}$. At each time slot, each user reads out its sequence value, and transmits when the sequence value is 1, otherwise keeps silent.

Remark 1: Note that in the sequence based scheme, we should perform sequence assignment at first so that each user schedules its transmissions according to its pre-assigned sequence. For systems in which the number of users is dynamic and extremely large, there are a variety of solutions proposed in the literature to reuse a sequence set of small size efficiently, such as the geographic location based sequence assignment in cellular networks [26] and the dynamic sequence assignment which leverages roadside nodes in vehicular ad hoc networks (VANETs) [27]. In this paper, our focus is on the design of schedule sequences and the analysis of their AoI performance, instead of the sequence assignment problem. Thus, we just assume the total number of users is known and design sequences based on it. When applying our sequence scheme to dynamic scenarios such as cellular networks and VANETs, we can also employ the sequence assignment methods proposed above to adapt to changes in the number of users and maintain good performance.

B. Synchronization Assumption on the Time Slotted Channel

We assume that there exists a system-wide reference time $t = 0$. The *starting time offset* of a user is defined as the time difference between the reference time and the time for this user to start its transmission session. We use τ_i to denote the starting time offset of user i , where $i \in \{1, 2, \dots, N\}$, as shown in Fig. 1. There are three different levels of time synchronization in slotted systems, as illustrated in Fig. 2.

- 1) Full synchronization: By accurate time synchronization, all users start their transmission session simultaneously.

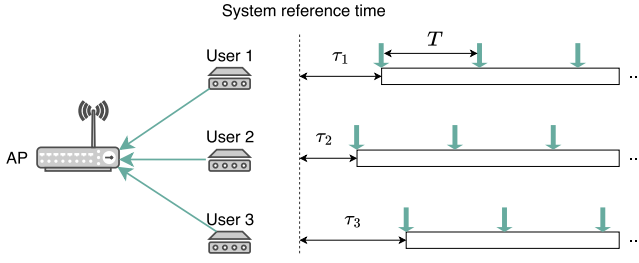


Fig. 1. Illustration of the system and arrival traffic model.

That is, the starting time offsets of all users are the same and can be set as all zeros without loss of generality. The frame boundaries of all users are aligned, as well as the slot boundaries.

- 2) Full asynchronization: Without time synchronization, each user starts its transmission session independently. It follows that the frame boundaries of all users are not necessarily aligned, nor are the slot boundaries. In this case, the starting time offsets of all users can be any real numbers.
- 3) Slot synchronization: The frame boundaries of all users are still unaligned, but the slot boundaries are aligned. The starting time offsets of all users can only assume values that are integral multiples of a slot duration.

It is challenging to achieve full synchronization which would cause heavy overhead, especially for energy-constrained systems. In view of the high cost for full synchronization, it is more desirable to design a fully asynchronous scheme. However, the sequence design in fully asynchronous cases, where the starting time offsets are unknown real numbers and transmissions from different users may partially overlap with each other, is much more complex. To facilitate discussions, we will focus on the slot-synchronous model since the results obtained in the slot-synchronous model can be easily extended to the fully asynchronous model, as demonstrated by prior studies [7], [28]. We will also show the AoI performance of the extended sequences in fully asynchronous model by simulations under Section VIII.

Without loss of generality, all users are assumed to start their schedules no later than $t = 0$. Thus, in the slot-synchronous case, the starting time offsets are non-negative integers. Since the sequences have period L , we assume that $\tau_i \in \mathbb{Z}_L$. We define $\boldsymbol{\tau} = (\tau_1, \tau_2, \dots, \tau_N) \in \mathbb{Z}_L^N$ to represent the starting time offsets for all the users.

C. Time Evolution of the AoI Process

The timeliness of the status updates is termed as the Age of Information (AoI), i.e., the time elapsed since the generation of the last successfully received status. We consider user i and let $A_i(t)$ be the instantaneous AoI of user i at the t -th time slot. As exemplified by Fig. 3, $A_i(t)$ grows linearly over time in the absence of successful update delivery from user i , i.e., $A_i(t) = A_i(t-1) + 1$. Conversely, as soon as a packet sent by user i is delivered successfully, the AP accordingly refreshes user i 's state and $A_i(t)$ drops to $A_i(t) = t - g_i(t)$, where $g_i(t)$ denotes the time at which this packet was generated.

As a result, we have

$$A_i(t) = \begin{cases} t - g_i(t), & \text{if a packet is successfully received at } t, \\ A_i(t-1) + 1, & \text{otherwise.} \end{cases} \quad (1)$$

Since the AP would not acknowledge for each packet transmission, the users have no way of knowing the result of each transmission and will not stop transmitting a packet even if a transmission of this packet has been successful, until its expiration. Therefore, a packet may be sent successfully for multiple times within a period. It should be pointed out that for each packet, only its first successful transmission would cause the instantaneous AoI to drop, while its subsequent successful transmissions after that would not. For user i , we denote the time slot when its instantaneous AoI experiences the j -th drop by d_i^j , $j = 1, 2, \dots$. For the packet successfully delivered at d_i^j , we define its service time S_i^j as the time interval between its generation time and d_i^j . Besides, we use Y_i^j to denote the inter-departure time between two consecutive successfully received updates, i.e.,

$$Y_i^j = d_i^{j+1} - d_i^j, j = 1, 2, \dots \quad (2)$$

Based on the AoI evolution, the average AoI (AAoI) of user i , denoted by $\bar{A}_i$, is defined as [29]

$$\bar{A}_i = \lim_{\lambda \rightarrow \infty} \frac{1}{\lambda} \sum_{t=0}^{\lambda} A_i(t). \quad (3)$$

Let $V(t) = \max_j \{j | d_i^j < t\}$ be the number of successfully transmitted statuses for user i until slot t , and let Q_i^j be the sum of AoI between d_i^j and d_i^{j+1} , as exemplified in the highlighted polygon area of Fig. 3. By the Renewal Reward Theorem [30], $\bar{A}_i$ is given by

$$\bar{A}_i = \lim_{t \rightarrow \infty} \frac{V(t)}{t} \frac{1}{V(t)} \sum_{j=1}^{V(t)} Q_i^j = \frac{\mathbb{E}[Q_i^j]}{\mathbb{E}[Y_i^j]}. \quad (4)$$

The area Q_i^j is calculated as $Q_i^j = S_i^j + (S_i^j + 1) + \dots + (S_i^j + Y_i^j - 1) = S_i^j Y_i^j + \frac{(Y_i^j)^2 - Y_i^j}{2}$. Therefore, the AAoI of user i is given by

$$\bar{A}_i = \frac{\mathbb{E}[Q_i^j]}{\mathbb{E}[Y_i^j]} = \frac{\mathbb{E}[S_i^j Y_i^j]}{\mathbb{E}[Y_i^j]} + \frac{\mathbb{E}[(Y_i^j)^2]}{2\mathbb{E}[Y_i^j]} - \frac{1}{2}. \quad (5)$$

We denote the j -th peak AoI of user i by A_i^j . It is given by $A_i^j = S_i^j + Y_i^j - 1$ [29]. Then, the average peak AoI (APAoI) of user i , denoted by $\bar{A}_i^p$, is derived as

$$\bar{A}_i^p = \mathbb{E}[S_i^j] + \mathbb{E}[Y_i^j] - 1. \quad (6)$$

Based on the instantaneous AoI of user i , $A_i(t)$, we further introduce *age penalty function* $g(A_i(t))$ to characterize the level of dissatisfaction on data staleness for user i . This function is usually defined depending on specific applications. For example, under an exponential age penalty function $g(A_i(t)) = e^{aA_i(t)}$ with $a > 0$, the larger the instantaneous AoI, the more rapidly the penalty increases. So, this function is suitable for real-time applications with extremely demanding

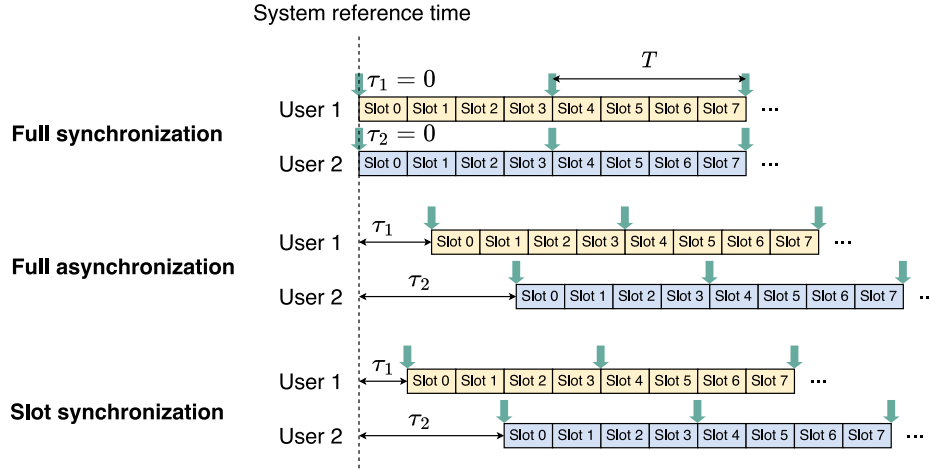


Fig. 2. Illustration of different synchronization levels.

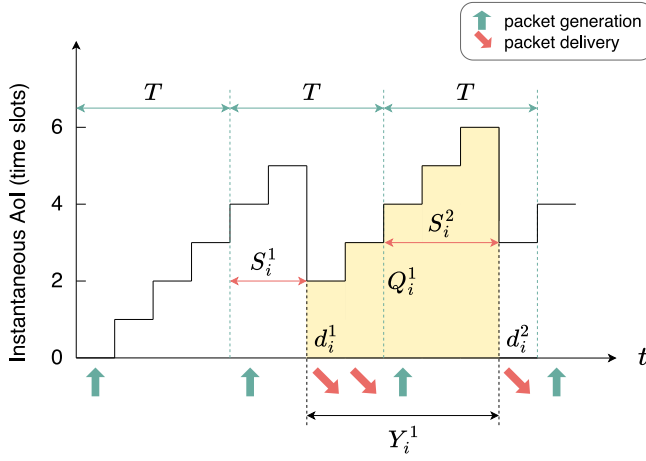
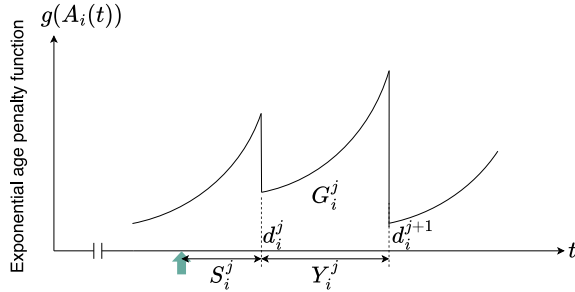
Fig. 3. The instantaneous AoI versus the index of time slot for user i .

Fig. 4. Time evolution of an exponential age penalty function.

requirements for status refreshing, such as online advertisement placement and online Web ranking [31]. The time evolution of an exponential age penalty function is illustrated in Fig. 4. We denote the sum of age penalty between drop d_i^j and its subsequent drop d_i^{j+1} by G_i^j , as shown in Fig. 4. Similar to the definition of $\bar{A}_i$, the average age penalty of user i , denoted by $\overline{g(A_i(t))}$, is given by

$$\overline{g(A_i(t))} = \frac{\mathbb{E}[G_i^j]}{\mathbb{E}[Y_i^j]}. \quad (7)$$

TABLE I
NOTATION TABLE

Notation	Definition
N	Total number of users
T	Period of status generation
L	Period of a schedule sequence set
τ_i	Starting time offset of user i , $\tau_i \in \mathbb{Z}_L$
τ	Combination of τ_i 's, i.e., $\tau = (\tau_1, \tau_2, \dots, \tau_N) \in \mathbb{Z}_L^N$
s_i	Schedule sequence assigned to user i
w_i	Hamming weight of s_i
$s_i^{\delta_i}$	Cyclic shift of s_i obtained by cyclically shifting s_i by δ_i
$\mathcal{C}_i$	Set of cyclic shifts of s_i
$\mathcal{S}$	$\mathcal{S} = \{s_1, s_2, \dots, s_N\}$
d_i^j	Time of the j -th drop for user i
S_i^j	Service time of the packet delivered at the j -th drop for user i
Y_i^j	Inter-departure time between the j -th and the $j+1$ -th drops for user i
$\bar{A}_i$	Average AoI (AAoI) of user i
$\bar{A}_i^P$	Average peak AoI (APAoI) of user i

IV. PRELIMINARIES

In this paper, we focus on AoI under a common class of schedule sequences, namely Minimum Hamming User-Irrepressible (MHUI) sequences. Analytical results on MHUI sequences will be derived in the coming sections. In this section, we introduce preliminaries to facilitate the derivation of analytical results, including necessary technical definitions and critical properties of MHUI sequences. Moreover, a most common construction method to build MHUI sequences is also provided. Besides, in order to facilitate the reading, we list the main notations used in this paper in Table I.

Definition 1: Hamming weight. The Hamming weight of a binary sequence is defined as the number of “1”s in it.

For s_i , $i \in \{1, 2, \dots, N\}$, its Hamming weight is denoted by w_i .

Definition 2: Characteristic set. For a binary sequence $s_i = [s_i(0) \ s_i(1) \ \dots \ s_i(L-1)]$, its characteristic set $\mathcal{I}_i$ is defined as a subset of $\mathbb{Z}_L$ such that $t \in \mathcal{I}_i$ if and only if $s_i(t) = 1$.

Definition 3: Hamming cross-correlation. For two binary sequences $s_1 = [s_1(0) \ s_1(1) \ \dots \ s_1(L-1)]$ and $s_2 = [s_2(0) \ s_2(1) \ \dots \ s_2(L-1)]$ with a common period L and

relative time offset τ , where $\tau \in \mathbb{Z}_L$, their Hamming cross-correlation is defined by $H_{1,2}(\tau) = \sum_{t=0}^{L-1} s_1(t)s_2(t \oplus_L \tau)$, where the addition $\oplus_L$ is performed in $\mathbb{Z}_L$.

Definition 4: Minimum Hamming cross-correlation property. For a set of N binary sequences of length L , the minimum Hamming cross-correlation property refers to that the Hamming cross-correlation between any two sequences in the set is no more than 1 for any starting time offsets, that is, $H_{i,j}(\tau) \leq 1$, for $i, j \in \{1, 2, \dots, N\}$, $i \neq j$, and $\tau \in \mathbb{Z}_L$.

Definition 5: MHUI sequence set. A set of N binary sequences is called a minimum Hamming user-irrepressible (MHUI) sequence set if it satisfies: 1) the Hamming weight of each sequence is no less than N ; and 2) the minimum Hamming cross-correlation property.

In this paper, we will focus on the AoI of MHUI sequence sets. According to Definition 5, each sequence out of an MHUI sequence set can be guaranteed to have at least one collision-free “1” within a sequence period, for any starting time offsets among them. Such a desirable property has attracted a lot of research interests in MHUI sequence sets. For example, an MHUI sequence set in which all sequences have the same Hamming weight is also known as a *conflict-avoiding code* (CAC) [32], which has been extensively studied in the literature.

Let $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$ be an MHUI sequence set of period L . For $s_i \in \mathcal{S}$ and $\delta_i \in \mathbb{Z}_L$, denote by $s_i^{\delta_i}$ the *cyclic shift* of s_i by δ_i , which is given by $s_i^{\delta_i} = [s_i(\delta_i) \ s_i(1 \oplus_L \delta_i) \ \dots \ s_i((L-1) \oplus_L \delta_i)]$, where $s_i = [s_i(0) \ s_i(1) \ \dots \ s_i(L-1)]$. Notice that the characteristic set of $s_i^{\delta_i}$ is $\{t \oplus_L \delta_i : t \in \mathcal{I}_i\}$. We denote the set of all L cyclic shifts of s_i by $\mathcal{C}_i$, i.e., $\mathcal{C}_i = \{s_i, s_i^1, s_i^2, \dots, s_i^{L-1}\}$. It is worth noting that, for $\delta_1, \delta_2, \dots, \delta_N \in \mathbb{Z}_L$, the set $\{s_1^{\delta_1}, s_2^{\delta_2}, \dots, s_N^{\delta_N}\}$ still forms an MHUI sequence set.

Remark 2: Note that δ_i is different from the starting time offset τ_i , $i \in \{1, 2, \dots, N\}$. The former is an adjustable parameter in sequence design or construction, while the latter is caused by the absence of time synchronization among the users and is unknown to the users.

The following CRT construction, based on the Chinese Remainder Theorem, can be used to produce an MHUI sequence set with a common period $L = p(2N-1)$, where p is the smallest prime satisfying that $p \geq N$.

Definition 6: CRT construction [23]: Given N , let Hamming weight $w_1 = w_2 = \dots = w_N = N$, $q = 2N-1$ and p be the smallest prime satisfying that $p \geq N$. For coprime p and q , we define CRT correspondence as a bijective mapping between $\mathbb{Z}_{pq}$ and $\mathbb{Z}_p \times \mathbb{Z}_q$: $\Phi_{p,q}(t) = (t \bmod p, t \bmod q)$, $t \in \mathbb{Z}_{pq}$. We construct a set of p sequences $\{s_g = [s_g(0) \ s_g(1) \ \dots \ s_g(L-1)] : g \in \{1, 2, \dots, p\}\}$ with a common period $L = pq$ as follows. For $t \in \mathbb{Z}_L$, we define

$$s_g(t) = \begin{cases} 1, & \text{if } \Phi_{p,q}(t) = (ug \bmod p, u \bmod q), \text{ for } u \in \mathbb{Z}_N, \\ 0, & \text{otherwise,} \end{cases}$$

where g is called the *generator* of s_g .

It is shown in [23] that the sequences based on CRT construction enjoy the minimum Hamming cross-correlation property, and thus form an MHUI sequence set.

Example 1: Let $N = 3$. We have $p = 3, q = 5$ and $L = 15$. The 3 resulting sequences based on the CRT construction are as follows.

$$\begin{aligned} s_1 &= [111000000000000], \\ s_2 &= [100000010001000], \\ s_3 &= [100000100000100]. \end{aligned}$$

One can check that the minimum Hamming cross-correlation property holds for any pair of $\{s_1^{\delta_1}, s_2^{\delta_2}, s_3^{\delta_3}\}$, for $\delta_1, \delta_2, \delta_3 \in \mathbb{Z}_{15}$.

V. AOI ANALYSIS FOR MHUI SEQUENCES

In this section, we analyze the AoI performance for MHUI sequences. Let $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$ be a set of N MHUI sequences of common period L and with Hamming weight w_i for s_i , $i = 1, 2, \dots, N$. Each user is pre-assigned a unique sequence from $\mathcal{S}$ to schedule transmissions of status packets. We consider here the case of $T = L$ where each user is guaranteed to transmit a status packet successfully at least once within a sequence period L (or equivalently, a status generation period T). It follows that for each user, the drop of instantaneous AoI must happen in every sequence period. It is worth noting that for any user, the positions of its AoI drops within every sequence period are the same. That is, for user i , the values of S_i^j 's are the same for different j . Then, the inter-departure time between any two consecutive drops d_i^{j+1} and d_i^j is given by $Y_i^j = L$, for $j = 1, 2, \dots$. Furthermore, we have $\mathbb{E}[Y_i^j] = L$ and $\mathbb{E}[(Y_i^j)^2] = L^2$. As a result, the AAoI of user i in (5) is simplified as

$$\overline{A_i} = \mathbb{E}[S_i^j] + \frac{L}{2} - \frac{1}{2}, \quad (8)$$

and the APAoI of user i in (6) is simplified as

$$\overline{A_i^p} = \mathbb{E}[S_i^j] + L - 1. \quad (9)$$

Next, we will consider how to derive $\mathbb{E}[S_i^j]$ for user i . We denote the characteristic set of s_i as $\mathcal{I}_i = \{x_{i,1}, x_{i,2}, \dots, x_{i,w_i}\}$, where $x_{i,1} < x_{i,2} < \dots < x_{i,w_i}$. Entries in $\mathcal{I}_i$ denote the time slot indices at which user i transmits its status packet within a sequence period. For each packet, the time for its first successful reception must be an entry of $\mathcal{I}_i$. We denote by $P_{i,k}$ the probability that the first successful reception happens at $x_{i,k}$, $k \in \{1, 2, \dots, w_i\}$. It follows that

$$\mathbb{E}[S_i^j] = \sum_{k=1}^{w_i} x_{i,k} P_{i,k}. \quad (10)$$

A closed form of $P_{i,k}$ is given as follows.

Lemma 1: Fix $i \in \{1, 2, \dots, N\}$. For each integer $n \leq N-1$, let $\binom{\mathcal{S} \setminus \{s_i\}}{n}$ denote the collection of all subsets of $\mathcal{S} \setminus \{s_i\}$ with size n . Then, for $k = 1, 2, \dots, w_i$, $P_{i,k}$ is given by (11), as shown at the bottom of the next page, where $S(n, k-1)$

denotes the *Stirling number of the second kind* [33]. When $w_1 = w_2 = \dots = w_N = w$, (11) can be simplified as

$$P_{i,k} = \frac{(k-1)!}{L^{N-1}} \sum_{n=k-1}^{N-1} \binom{N-1}{n} S(n, k-1) w^n (L - kw)^{N-1-n}. \quad (12)$$

Proof: To calculate $P_{i,k}$, we count the number of starting time offsets τ 's which lead to the result that the transmissions at $x_{i,1}, x_{i,2}, \dots, x_{i,k-1}$ fail and the transmission at $x_{i,k}$ succeeds. The counting steps for this number, denoted by N_τ , are presented as follows.

- 1) Without loss of generality, we fix s_i and shift the other $(N-1)$ sequences in $\mathcal{S} \setminus \{s_i\}$. We should find a subset of $\mathcal{S} \setminus \{s_i\}$ and shift the sequences in the subset to collide the first $(k-1)$ "1"s in s_i . Due to the minimum Hamming cross-correlation property, each sequence can only block one "1" in s_i , no matter what the starting time offset between them is. Therefore, the size of the subset, denoted by n , should be no less than $(k-1)$. Thus, we have $n \in \{k-1, k, \dots, N-1\}$.
- 2) For an n -subset $\mathcal{B} \in \binom{\mathcal{S} \setminus \{s_i\}}{n}$, we shall count the number of ways for $\mathcal{B}$ to collide the first $(k-1)$ "1"s in s_i , under the condition that each sequence can collide at most one "1" in s_i . This is equivalent to partitioning a set of n distinct balls into $(k-1)$ labeled boxes such that no box is empty. The number of such partitions is $\sum_{r=0}^{k-1} (-1)^r \binom{k-1}{r} (k-1-r)^n$, which is equal to $(k-1)! S(n, k-1)$.
- 3) For $s_j \neq s_i$, there are exactly w_j starting time offsets so that it can block a particular "1" in s_i . Since the offset of each sequence is independent of the other, the total number of combinations of offsets for these n sequences in $\mathcal{B}$ is $\prod_{s_j \in \mathcal{B}} w_j$. For $s_j \neq s_i$, there are exactly $(L - kw_j)$ starting time offsets so that it has no collisions with the first k "1"s in s_i . Then, the total number of combinations of offsets for sequences in $\mathcal{S} \setminus (\mathcal{B} \cup \{s_i\})$ is $\prod_{s_j \in \mathcal{S} \setminus (\mathcal{B} \cup \{s_i\})} (L - kw_j)$.

By combining the above counting steps, the number of τ 's that lead to the first successful transmission to occur at $x_{i,k}$, N_τ , is given by (13), as shown at the bottom of the page. Since the total number of τ 's of the $(N-1)$ sequences in $\mathcal{S} \setminus \{s_i\}$ is L^{N-1} , the probability $P_{i,k}$ is given by N_τ / L^{N-1} . This completes the proof. ■

Based on Lemma 1, we derive $\lim_{N \rightarrow \infty} P_{i,k}$ as below.

TABLE II

LIMIT OF $P_{i,k}$ FROM $k = 1$ TO $k = 10$ (ROUNDED TO SIX DECIMAL PLACES)

$\lim_{N \rightarrow \infty} P_{i,1}$	0.606531	$\lim_{N \rightarrow \infty} P_{i,6}$	0.00572
$\lim_{N \rightarrow \infty} P_{i,2}$	0.238651	$\lim_{N \rightarrow \infty} P_{i,7}$	0.002251
$\lim_{N \rightarrow \infty} P_{i,3}$	0.093902	$\lim_{N \rightarrow \infty} P_{i,8}$	0.000886
$\lim_{N \rightarrow \infty} P_{i,4}$	0.036948	$\lim_{N \rightarrow \infty} P_{i,9}$	0.000348
$\lim_{N \rightarrow \infty} P_{i,5}$	0.014538	$\lim_{N \rightarrow \infty} P_{i,10}$	0.000137

Theorem 2: For MHUI sequences under CRT construction, we have

$$\lim_{N \rightarrow \infty} P_{i,k} = \left(1 - e^{-\frac{1}{2}}\right)^{k-1} e^{-\frac{1}{2}}. \quad (14)$$

The proof is given in Appendix A. We list $\lim_{N \rightarrow \infty} P_{i,k}$ from $k = 1$ to $k = 10$ in Table II, which are obtained according to Theorem 2. Lemma 1 and Theorem 2 provide the following two essential insights for $P_{i,k}$.

- 1) Lemma 1 indicates that $P_{i,k}$ is independent of the position of "1"s in s_i , since no term involving position information of "1"s appears in the closed-form expression of $P_{i,k}$.
- 2) Theorem 2 indicates that $\lim_{N \rightarrow \infty} P_{i,k}$ decreases monotonically with regard to k , which means that the earlier the transmission in a sequence period, the more likely it will be the first successful transmission within that period. Moreover, we can observe from Table II that entries with small k dominate the table. Especially, $\lim_{N \rightarrow \infty} P_{i,1}$ is much larger than the other entries.

After obtaining results on $P_{i,k}$, we combine (10) and (11) to derive the closed-form expression for $\mathbb{E}[S_i^j]$ as follows.

Lemma 3: For user i assigned with s_i , $i \in \{1, 2, \dots, N\}$, $\mathbb{E}[S_i^j]$ is given by (15), as shown at the bottom of the next page.

We learn an important insight from Lemma 3 that $\mathbb{E}[S_i^j]$ only depends on the distribution of "1"s in s_i and is independent of the distribution of "1"s in other users' sequences. Thus, changing the distribution of "1"s in a sequence will not affect the performance of other sequences.

Based on (8), (9) and Lemma 3, the AAOI and the APAOI of user i are obtained as follows.

Theorem 4: For user i assigned with s_i , $i \in \{1, 2, \dots, N\}$, $\bar{A}_i$ and $\bar{A}_i^p$ are given by (16) and (17), as shown at the bottom of the next page, respectively. When $w_1 = w_2 = \dots = w_N = w$, (16) and (17) can be simplified as (18) and (19), as shown at the bottom of the next page, respectively.

$$P_{i,k} = \frac{(k-1)!}{L^{N-1}} \sum_{n=k-1}^{N-1} S(n, k-1) \sum_{\mathcal{B} \in \binom{\mathcal{S} \setminus \{s_i\}}{n}} \left(\prod_{s_j \in \mathcal{B}} w_j \right) \left(\prod_{s_j \in \mathcal{S} \setminus (\mathcal{B} \cup \{s_i\})} (L - kw_j) \right). \quad (11)$$

$$N_\tau = (k-1)! \sum_{n=k-1}^{N-1} S(n, k-1) \sum_{\mathcal{B} \in \binom{\mathcal{S} \setminus \{s_i\}}{n}} \left(\prod_{s_j \in \mathcal{B}} w_j \right) \left(\prod_{s_j \in \mathcal{S} \setminus (\mathcal{B} \cup \{s_i\})} (L - kw_j) \right). \quad (13)$$

For the average age penalty of user i , we have

$$\overline{g(A_i(t))} = \frac{\mathbb{E}[G_i^j]}{\mathbb{E}[Y_i^j]} = \sum_{k=1}^{w_i} P_{i,k} \sum_{t=x_{i,k}}^{x_{i,k}+L-1} \frac{g(A_i(t))}{L}. \quad (20)$$

Based on Lemma 1, the average age penalty of user i can be obtained as follows.

Theorem 5: For user i assigned with s_i , $i \in \{1, 2, \dots, N\}$, $\overline{g(A_i(t))}$ is given by (21), as shown at the bottom of the next page. When $w_1 = w_2 = \dots = w_N = w$, (21) can be simplified as (22), shown at the bottom of the next page.

Theorem 5 is useful for analyzing system performance under different age penalty functions, for sequence-based scheduling. One of our benchmarks, slotted ALOHA, also has a closed-form expression for AoI, which is given in [11] and can be used to calculate the average age penalty. However, there have been no previous studies on the age penalty under sequence based scheme. Our results in Theorem 5 fill the gap in the literature.

VI. OPTIMIZATION FOR MHUI SEQUENCES

The AoI performance of MHUI sequences in the case of $T = L$ has been derived analytically in Section V. The results are under the premise that the MHUI sequence set $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$ is specified. In this section, we consider how to optimize $\mathcal{S}$ with the objective of minimizing AoI under $\mathcal{S}$. We focus on the average service time, since the inter-departure time keeps equal to L .

We will optimize $\mathcal{S}$ by choosing the optimal cyclic shift for each sequence in $\mathcal{S}$ in an efficient way. Our motivations for doing so are as follows.

- 1) The importance of optimal cyclic shift. We learn from Lemma 3 that the average service time of user i , $i \in \{1, 2, \dots, N\}$, which is associated with s_i , depends on the characteristic set of s_i . For each sequence, choosing

different cyclic shifts would result in potentially different characteristic sets. Thus, choosing an optimal cyclic shift from $\mathcal{C}_i$ for each s_i is crucial for minimizing the average service time of user i , where $\mathcal{C}_i$ denotes the set of all cyclic shifts of s_i , i.e., $\mathcal{C}_i = \{s_i, s_i^1, s_i^2, \dots, s_i^{L-1}\}$. The example below illustrates this point.

Example 2: Consider a cyclic shift of s_1 in Example 1: $s_1^5 = [0 0 0 0 0 0 0 0 0 1 1 1 0 0]$ with characteristic set $\{10, 11, 12\}$. Both s_1 and s_1^5 guarantee that user 1 can transmit at least one packet successfully within a sequence period. However, s_1 and s_1^5 lead to quite different AoI performance. By using s_1 , the service time of user 1 is in the range of $\{0, 1, 2\}$, while by using s_1^5 , the range of service time is much higher, that is, $\{10, 11, 12\}$.

- 2) The complexity to find optimal cyclic shift. It is worth emphasizing that according to Lemma 3, the AoI performance of user i is independent of the characteristic sets of other users' sequences. Therefore, we can optimize the cyclic shift of each s_i separately. The optimal cyclic shift of s_i is denoted by s_i^* . For each s_i , a trivial idea to find s_i^* is to go through all the L cyclic shifts in $\mathcal{C}_i$, among which s_i^* is the one that yields the shortest $\mathbb{E}[S_i^j]$. However, going through all L cyclic shifts is time-consuming, and thus a more efficient method is desirable.

In the following, we will elaborate on how to obtain s_i^* more efficiently. We learn from Lemma 1 that $P_{i,k}$ is independent of the characteristic set of s_i , or equivalently, the cyclic shift of s_i . Based on this observation, we have the following property on s_i^* .

Theorem 6: Given an MHUI sequence s_i , if $s_i^* = [s_i^*(0) s_i^*(1) \dots s_i^*(L-1)]$ is its optimal cyclic shift, then we must have $s_i^*(0) = 1$.

$$\mathbb{E}[S_i^j] = \sum_{k=1}^{w_i} \frac{x_{i,k}(k-1)!}{L^{N-1}} \sum_{n=k-1}^{N-1} S(n, k-1) \sum_{\mathcal{B} \in (\mathcal{S} \setminus \{s_i\})} \left(\prod_{s_j \in \mathcal{B}} w_j \right) \left(\prod_{s_j \in \mathcal{S} \setminus (\mathcal{B} \cup \{s_i\})} (L - kw_j) \right). \quad (15)$$

$$\overline{A_i} = \sum_{k=1}^{w_i} \frac{x_{i,k}(k-1)!}{L^{N-1}} \sum_{n=k-1}^{N-1} S(n, k-1) \sum_{\mathcal{B} \in (\mathcal{S} \setminus \{s_i\})} \left(\prod_{s_j \in \mathcal{B}} w_j \right) \left(\prod_{s_j \in \mathcal{S} \setminus (\mathcal{B} \cup \{s_i\})} (L - kw_j) \right) + \frac{L-1}{2}. \quad (16)$$

$$\overline{A_i^p} = \sum_{k=1}^{w_i} \frac{x_{i,k}(k-1)!}{L^{N-1}} \sum_{n=k-1}^{N-1} S(n, k-1) \sum_{\mathcal{B} \in (\mathcal{S} \setminus \{s_i\})} \left(\prod_{s_j \in \mathcal{B}} w_j \right) \left(\prod_{s_j \in \mathcal{S} \setminus (\mathcal{B} \cup \{s_i\})} (L - kw_j) \right) + L - 1. \quad (17)$$

$$\overline{A_i} = \sum_{k=1}^{w_i} \frac{x_{i,k}(k-1)!}{L^{N-1}} \sum_{n=k-1}^{N-1} \binom{N-1}{n} S(n, k-1) w^n (L - kw)^{N-1-n} + \frac{L-1}{2}. \quad (18)$$

$$\overline{A_i^p} = \sum_{k=1}^{w_i} \frac{x_{i,k}(k-1)!}{L^{N-1}} \sum_{n=k-1}^{N-1} \binom{N-1}{n} S(n, k-1) w^n (L - kw)^{N-1-n} + L - 1. \quad (19)$$

Algorithm 1 Algorithm for Optimizing $\mathcal{S}$ **Require:** A set of MHUI sequences $\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_N$

- 1: **for** each $i \in \{1, 2, \dots, N\}$ **do**
- 2: calculate $\mathcal{C}_i^1$ for $\mathbf{s}_i$;
- 3: **for** each cyclic shift in $\mathcal{C}_i^1$, calculate $\mathbb{E}[S_i^j]$ based on Lemma 3;
- 4: choose the cyclic shift $\mathbf{s}_i^*$ which corresponds to the smallest $\mathbb{E}[S_i^j]$;
- 5: **end for**

Ensure: $\mathcal{S} = \{\mathbf{s}_1^*, \mathbf{s}_2^*, \dots, \mathbf{s}_N^*\}$

Proof: This can be proved by contradiction. Assume that $\mathbf{s}_i^*$ is the optimal cyclic shift of $\mathbf{s}_i$, and $\mathbf{s}_i^*(0) = 0$. Denote the characteristic set of $\mathbf{s}_i^*$ by $\mathcal{I}_i = \{x_{i,1}, x_{i,2}, \dots, x_{i,w_i}\}$, where $x_{i,1} < x_{i,2} < \dots < x_{i,w_i}$. The average service time of user i under $\mathbf{s}_i^*$ is assumed to be the smallest among all the cyclic shifts in $\mathcal{C}_i$. It is given by $\mathbb{E}[S_i^j | \mathbf{s}_i^*] = \sum_{k=1}^{w_i} x_{i,k} P_{i,k}$. Since $\mathbf{s}_i^*(0) = 0$, we have $x_{i,1} > 0$. Let $\mathbf{s}_i'$ be the cyclic shift obtained by cyclically shifting $\mathbf{s}_i^*$ by $x_{i,1}$. Then, the characteristic set of $\mathbf{s}_i'$ is $\mathcal{I}_i' = \{0, x_{i,2} - x_{i,1}, \dots, x_{i,w_i} - x_{i,1}\}$. It follows that the average service time under $\mathbf{s}_i'$ equals $\mathbb{E}[S_i^j | \mathbf{s}_i'] = \sum_{k=1}^{w_i} (x_{i,k} - x_{i,1}) P_{i,k}$. Obviously, $\mathbb{E}[S_i^j | \mathbf{s}_i'] < \mathbb{E}[S_i^j | \mathbf{s}_i^*]$. Thus, the assumption that $\mathbf{s}_i^*(0) = 0$ cannot hold. This completes the proof. ■

Theorem 6 reveals the importance to choose a cyclic shift such that a “1” is allocated in the position of $t = 0$ in this cyclic shift. Moreover, according to Table II which is obtained by Theorem 2, $\lim_{N \rightarrow \infty} P_{i,1}$ is much larger than the other entries, which also implies the significance to allocate a “1” in the beginning of each sequence. For each $\mathbf{s}_i$, we denote $\mathcal{C}_i^1$ as the set of its cyclic shifts in each of which a “1” appears in the position of $t = 0$. Inspired by Theorem 6, we propose Algorithm 1 to find $\mathbf{s}_i^*$. Algorithm 1 improves the search efficiency for $\mathbf{s}_i^*$, since the search range is narrowed down from $\mathcal{C}_i$ to $\mathcal{C}_i^1$ which only contains w_i elements. That is, the search efficiency for $\mathbf{s}_i^*$ is enhanced by $(L/w_i)\times$.

Example 3: We have checked the effectiveness of Algorithm 1 in the case of $N = 8$. A set of 11 sequences $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_{11}\}$ with each of length 165 and Hamming weight 8, is given by the CRT construction. For each sequence $\mathbf{s}_i$, it has 165 cyclic shifts in total in $\mathcal{C}_i$, and among them, 8 cyclic shifts belong to $\mathcal{C}_i^1$, $i \in \{1, 2, \dots, 11\}$. As shown in Fig. 5, we have depicted the maximum average service time (indicated by the blue diamond markers) and the minimum average service time (indicated by the blue circle markers) under all the 165 cyclic shifts in $\mathcal{C}_i$, and the maximum

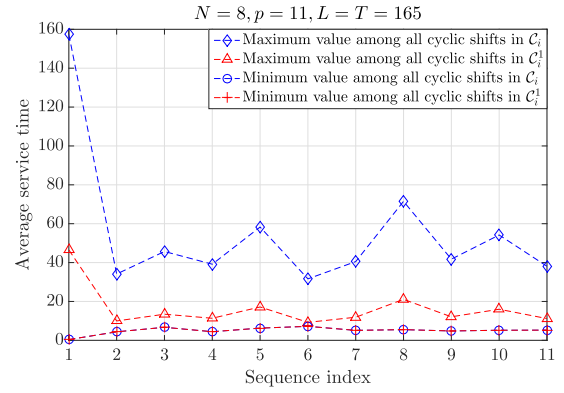


Fig. 5. Average service time for different cyclic shifts of $\mathbf{s}_i$, $i \in \{1, 2, \dots, 11\}$.

TABLE III

THE TOP 8 SEQUENCES WITH SMALLEST AVERAGE SERVICE TIME (ROUNDED TO TWO DECIMAL PLACES)

No.	Sequence	$\mathbb{E}[S_i^j]$
1	$\mathbf{s}_1^0$	0.39
2	$\mathbf{s}_2^8$	4.38
3	$\mathbf{s}_3^{38}$	4.44
4	$\mathbf{s}_4^{10}$	4.82
5	$\mathbf{s}_5^{83}$	5.13
6	$\mathbf{s}_6^{39}$	5.19
7	$\mathbf{s}_7^{11}$	5.25
8	$\mathbf{s}_8^{98}$	5.50

average service time (indicated by the red triangle markers) and the minimum average service time (indicated by the red plus markers) under the 8 cyclic shifts in $\mathcal{C}_i^1$, for each sequence $\mathbf{s}_i$. The values noted by the blue diamond markers are significantly higher than those noted by the red triangle markers. This demonstrates the importance to allocate a “1” in the first position of each sequence. Moreover, the blue circle markers are in line with the red plus markers. Therefore, when we search for $\mathbf{s}_i^*$, we only need to focus on $\mathcal{C}_i^1$ instead of $\mathcal{C}_i$, and thus the search efficiency for $\mathbf{s}_i^*$ can be improved by 20.625x. The top 8 cyclic shifts with the smallest average service time are listed in Table III below, which will be assigned to the 8 users. That is,

$$\mathcal{S} = \{\mathbf{s}_1^0, \mathbf{s}_2^8, \mathbf{s}_3^{38}, \mathbf{s}_4^{10}, \mathbf{s}_5^{83}, \mathbf{s}_6^{39}, \mathbf{s}_7^{11}, \mathbf{s}_8^{98}\}. \quad (23)$$

VII. EXTENDED DISCUSSIONS FOR THE CASE OF $T \neq L$

The discussions in previous sections are based on the assumption of $T = L$. In this section, we consider the case where the status generation period T may be different from L .

$$\overline{g(A_i(t))} = \sum_{k=1}^{w_i} \frac{(k-1)!}{L^{N-1}} \sum_{n=k-1}^{N-1} S(n, k-1) \sum_{\mathbf{B} \in (\mathcal{S} \setminus \{\mathbf{s}_i\})} \left(\prod_{\mathbf{s}_j \in \mathbf{B}} w_j \right) \left(\prod_{\mathbf{s}_j \in \mathcal{S} \setminus (\mathbf{B} \cup \{\mathbf{s}_i\})} (L - kw_j) \right)^{x_{i,k} + L - 1} \sum_{t=x_{i,k}} \frac{g(A_i(t))}{L}. \quad (21)$$

$$\overline{g(A_i(t))} = \sum_{k=1}^{w_i} \frac{(k-1)!}{L^{N-1}} \sum_{n=k-1}^{N-1} S(n, k-1) \binom{N-1}{n} S(n, k-1) w^n (L - kw)^{N-1-n} \sum_{t=x_{i,k}}^{x_{i,k} + L - 1} \frac{g(A_i(t))}{L}. \quad (22)$$

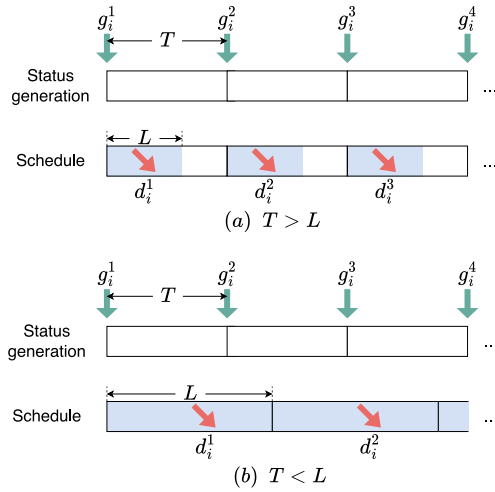


Fig. 6. Transmission schedules for user i in the cases of $T > L$ and $T < L$, respectively. The blue blocks of length L represent the assigned sequences, while the upside-down arrows represent the periodically generated statuses and the slanted arrows represent the successfully delivered statuses. In this example, each sequence can guarantee exactly one successful packet transmission within a period L . In the case of $T > L$, as shown in (a), each status g_i^j can be successfully delivered at d_i^j within L . However, for the case of $T < L$, as shown in (b), we can find that statuses g_i^1 and g_i^3 can be successfully delivered at d_i^1 and d_i^2 , respectively, while status g_i^2 cannot be successfully delivered.

We still pre-design a set of N MHUI sequences of length L and assign user i the sequence s_i , $i \in \{1, 2, \dots, N\}$, as in the case of $T = L$. If $T > L$, it is obvious that each user can still be guaranteed to send its up-to-date status successfully at least once within T . In this case, within each T , each user just transmits its status according to its assigned sequence in the first L slots, and keeps silent in the remaining $(T - L)$ slots. However, if $T < L$, this guarantee may not hold within T . For this case, we just let each user periodically repeat its assigned sequence, no matter what T is. In this way, the MHUI sequences still guarantee that each user can transmit a status packet successfully at least once within L , although some statuses may fail to be delivered. Fig. 6(a) and Fig. 6(b) below illustrate the sequence based transmission schedules for user i in the cases of $T > L$ and $T < L$, respectively.

We consider here the AoI performance under the sequence scheme when $T \neq L$. The analysis for the $T > L$ case can be conducted by extending the results under the case of $T = L$. We now focus on the case of $T < L$, which is more challenging. Under several special cases of $T < L$, closed-form expressions on the AoI performance have been derived. However, these results are omitted in this paper due to a shortage of space. Instead, we will briefly present how to determine the optimal cyclic shift for s_i in this case.

Recall that when $T = L$, the instantaneous AoI of user i changes periodically with period L . Let β be the least common multiple (LCM) of T and L . It is obvious that in the case of $T < L$, the instantaneous AoI of user i changes periodically with period β . We restrict our attention within a superframe of length β , which consists of β/T status generation periods. The schedule sequence of user i within a superframe is denoted

by $\mathbf{r}_i = \underbrace{[s_i \dots s_i]}_{\beta/L}$. As in the case of $T = L$, the AoI

performance of user i only depends on the characteristic set of s_i , and is independent of the characteristic sets of other users' sequences. Thus, the cyclic shift of each s_i can be optimized separately. In the following, we will present an efficient search method to find the optimal cyclic shift for s_i , based on the periodicity property described in Proposition 7 below.

Proposition 7: Denote α as the greatest common divisor (GCD) of T and L . Given $\mathbf{r}_i = \underbrace{[s_i \dots s_i]}_{\beta/L}$, the

AoI performance of user i within a superframe under $\mathbf{r}_i^{\delta_{i,1}} = \underbrace{[s_i^{\delta_{i,1}} \dots s_i^{\delta_{i,1}}]}_{\beta/L}$ is the same as that under $\mathbf{r}_i^{\delta_{i,2}} = \underbrace{[s_i^{\delta_{i,2}} \dots s_i^{\delta_{i,2}}]}_{\beta/L}$ if $\delta_{i,2} \equiv \delta_{i,1} \pmod{\alpha}$, where $\delta_{i,1}, \delta_{i,2} \in \mathbb{Z}_L$.

The proof of Proposition 7 is given in Appendix B. According to Proposition 7, the AoI performance of user i under $\mathbf{r}_i^{\delta_i}$ changes periodically with period α as δ_i goes through $\mathbb{Z}_L$. Benefiting from this periodicity property, when we search for the optimal value of δ_i , the search range can be narrowed down from $\mathbb{Z}_L$ to $\mathbb{Z}_\alpha$. That is, the search efficiency can be improved by $(L/\alpha)x$. Specifically, if $\alpha = 1$, which means that T is co-prime with L , then any cyclic shift of s_i would lead to the same AoI performance.

Example 4: We consider the case of $N = 5$, where a set of N MHUI sequences $\{s_1, s_2, \dots, s_5\}$ of $L = 45$ is obtained by CRT construction. As shown in Fig. 7, two values of T have been considered: $T = 8$ and $T = 15$. When $T = 8$, we have $\alpha = \text{GCD}(T, L) = 1$ and $\beta = \text{LCM}(T, L) = 360$. When $T = 15$, we have $\alpha = \text{GCD}(T, L) = 15$ and $\beta = \text{LCM}(T, L) = 45$. Fig. 7 depicts the AAoI of user 5 under sequence $\mathbf{r}_5^{\delta_5} = \underbrace{[s_5^{\delta_5} \dots s_5^{\delta_5}]}_{\beta/L}$ with δ_5 going through $\mathbb{Z}_{45}$, within a superframe of length β , for both $T = 8$ and $T = 15$. We observe from Fig. 7 that when $T = 15$, the AAoI changes periodically with period $\alpha = 15$. For example, for the cyclic shifts with $\delta_5 = 13, 28$ and 43 , the corresponding AAoI values are the same. The AAoI value here is the largest among all the values, which is 46% higher than the smallest AAoI achieved when $\delta_5 = 4, 19$ and 34 . This significant gap highlights the importance of choosing an optimal value for δ_5 . When $T = 8$, as indicated by the red circle markers, taking different δ_5 in $\mathbb{Z}_L$ makes no difference to the AAoI performance. That is, any cyclic shift of s_5 has the same AAoI in this case.

VIII. SIMULATION RESULTS

In this section, we present computer simulation results on AoI performance under various transmission scheduling schemes. We compare MHUI sequence scheme with two baseline schemes: first, the slotted ALOHA scheme, in which at each time slot each of the N users conducts transmission with a fixed probability p_t and remains idle with probability $(1 - p_t)$; second, the framed ALOHA scheme, in which at the beginning of each status generation period, each of the N users randomly picks w_{FA} slots for transmission,

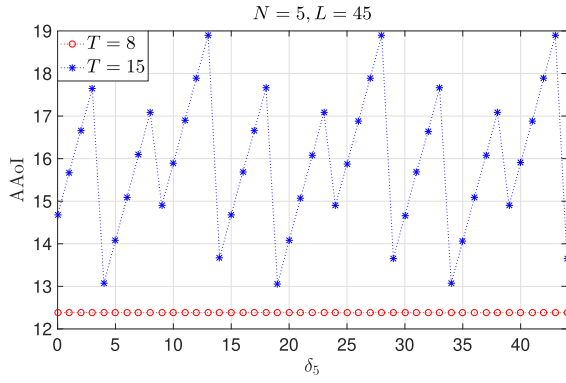


Fig. 7. Illustration for the periodicity of AoI performance under $\tau_5^{\delta_5}$.

$w_{FA} \in \{1, 2, \dots, T\}$. In each of the two schemes, one can enhance the AoI performance by optimizing p_t (resp., w_{FA}). During the comparison, we also take the duty factor of each scheme into consideration, which is defined as the ratio of transmission slots within a status generation period. The duty factor is an indicator of energy consumption. Specifically, higher duty factor represents higher energy consumption.

We first plot our derived closed-form expressions for the MHUI sequence scheme against the simulation results to demonstrate their correctness in Subsection VIII-A. Next, from Subsection VIII-B to Subsection VIII-F, we compare the performance of the sequence scheme and the baselines under various settings. In Subsection VIII-B and Subsection VIII-C, the comparisons are conducted in both slot-synchronous and asynchronous scenarios under the assumption of $T = L$. In Subsection VIII-D, comparison results on AoI penalty are presented. The comparisons in the unreliable channel case are shown in Subsection VIII-E. Furthermore, in Subsection VIII-F, we show the AoI performance under the consideration of $T < L$. For each simulation experiment, we simulate the evolution of AoI in 10^6 runs to obtain the average value. The starting time offset of each user in each run is randomly generated.

In MHUI sequence scheme, the deployed sequence set $\mathcal{S}$ is obtained by CRT construction, and is optimized according to our proposed Algorithm 1. The duty factor of each user in the MHUI sequence scheme is given by $f_s = N/L$. For the two baseline schemes, we also optimize them to minimize their AoI, as follows. For the slotted ALOHA scheme, by extending the proof in [11], the probability for each user to transmit a packet at a time slot is optimized at $p_t^* = 1/N$, which is also the average duty factor of the optimal slotted ALOHA scheme, denoted by $f_{SA}^* = 1/N$. For the framed ALOHA scheme, we determine its optimal number of transmission attempts for each status, denoted by w_{FA}^* , by exhaustive search in the range $[1, T]$. It follows that the optimal duty factor of the framed ALOHA is given by $f_{FA}^* = w_{FA}^*/T$. Note that the optimal duty factors of the slotted ALOHA and framed ALOHA schemes are usually higher than f_s . Thus, we also study the performance of the two baseline schemes with duty factor same as f_s in parallel. We label the above baseline schemes as “Optimal SA”, “Same SA”, “Optimal FA” and

TABLE IV
PARAMETERS FOR SCHEMES UNDER COMPARISON

Scheme	Duty factor	Hamming weight
MHUI sequence	$f_s = N/L$	$w = N$
Optimal SA	$f_{SA}^* = 1/N$	—
Same SA	$f_{SA}^s = N/L$	—
Optimal FA	$f_{FA}^* = w_{FA}^*/T$	w_{FA}^* obtained by exhaustive search
Same FA	$f_{FA}^s = N/L$	$w_{FA}^s = f_{FA}^s * T$

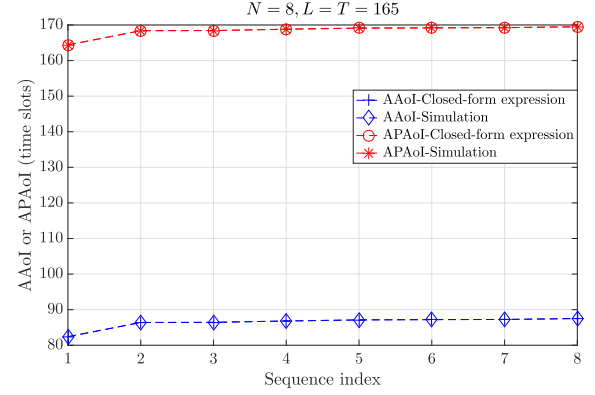


Fig. 8. The closed-form expressions against simulation results for AAoI and APAoI under the MHUI sequence scheme.

“Same FA” for short, and summarize the parameters of these schemes in Table IV.

A. Comparison of Closed-Form Expression against Simulation Result for MHUI Sequences

We have derived the closed-form expressions of AAoI and APAoI under the MHUI sequence scheme in Section V. Here, we plot them and compare them with simulation results. We consider the case of $N = 8$ and assign the sequence set $\mathcal{S}$ indicated by (23) to the 8 users. Fig. 8 depicts the AAoI and APAoI for each sequence in (23). We can see that the closed-form expressions for AAoI and APAoI are consistent with the simulation results.

B. AoI Performance in Slot-Synchronous Scenario

In the following, we compare AoI performance of the MHUI sequence scheme and the baseline schemes under the slot-synchronous scenario. We consider $N = 8$. In the MHUI sequence scheme, the sequences assigned to the 8 users follow (23) and have length $L = 165$ and duty factor $f_s = N/L = 0.0485$. The status generation period T is set as $T = L$. In the Optimal SA scheme, the optimal duty factor $f_{SA}^* = p_t^* = 1/N = 0.125$. For the Optimal FA scheme, we have the optimal $w_{FA}^* = 20$ by exhaustive search, and thus $f_{FA}^* = w_{FA}^*/T = 0.1212$. Note that f_{FA}^* is close to f_{SA}^* , and both of them are higher than f_s . This implies that employing either the Optimal SA scheme or the Optimal FA scheme costs more energy than the MHUI sequence scheme. We also simulate the Same SA and Same FA schemes which have the same duty factor as the MHUI sequence scheme. For the Same FA scheme, the number of transmission attempts for each status is set as $w_{FA}^s = 8$.

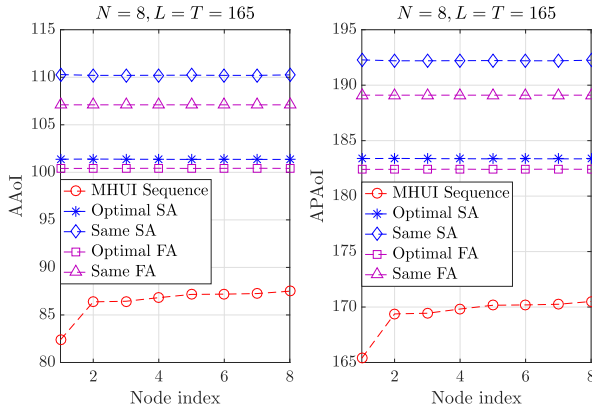


Fig. 9. AoI performance under the MHUI sequence scheme and the baselines in slot-synchronous scenario.

Fig. 9 shows the AAOI and APAOI performance of the various schemes. We can see that the MHUI sequence scheme outperforms the Optimal SA and Optimal FA schemes on both AAOI and APAOI. Note that the performance of the Optimal SA and Optimal FA schemes is similar, where the latter performs slightly better than the former. We take AAOI as an example. The average AAOI of the 8 users under the Optimal SA scheme (resp., Optimal FA scheme) is 101.4 (resp., 100.4), which is 17.4% (resp., 16.2%) higher than that under the MHUI sequence scheme, 86.4. Under the Same SA scheme (resp., Same FA scheme) which has the same energy assumption as the MHUI sequence scheme, the average AAOI of the 8 users is 110.2 (resp., 107.1), which is 27.5% (resp., 24%) higher than that under the MHUI sequence scheme.

C. AoI Performance in Asynchronous Scenario

Note that in this paper, we mainly focus on the slot-synchronous case in which the slot boundaries of each user are aligned and thus the starting time offsets are integers. This slot-synchronous case is a simplified version of the asynchronous case in which the slot boundaries are not aligned and thus the starting time offsets can be arbitrary real values. In this subsection, we extend the MHUI sequences obtained for the slot-synchronous case to the asynchronous case by the most common extension method in the literature [7], that is, padding a “0” after each entry in each sequence, as illustrated in Fig. 10. As a result, the newly obtained sequence has length $L' = 2L$. This extension method guarantees that each user can transmit a packet successfully at least once within L' . Fig. 11 compares the AAOI and APAOI performance of the various schemes. In the simulation, we have $N = 8$, $L = 165$, $L' = 2L = 330$, and the status generation period T is set as $T = L' = 330$. The sequences assigned to the 8 users are the extended versions of the sequences indicated in (23). For the baselines, we also employ the padding “0” scheme, since padding “0” also improves their AoI performance according to extensive simulations. Noting that by saying padding “0”, the meaning is slightly different in the baselines compared with the sequence scheme. Specifically, for the slotted ALOHA scheme, padding “0” means transmitting with a certain probability (f_{SA}^* or f_{FA}^s) in each of the odd-numbered slots within

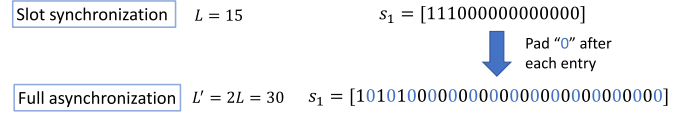


Fig. 10. Illustration for the extension method.

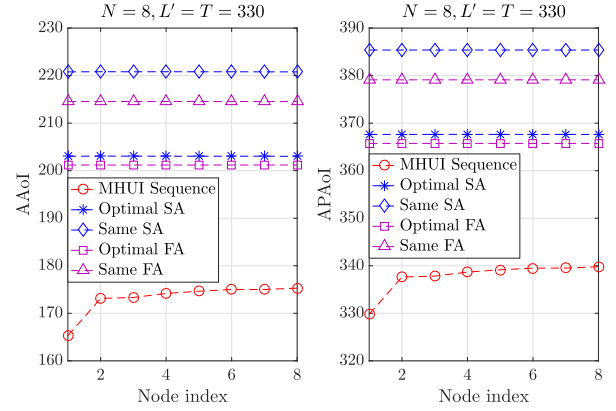


Fig. 11. AoI performance under the MHUI sequence scheme and the baselines in asynchronous scenario.

a period $T = L'$ and keeping silent in all the other slots. For the framed ALOHA scheme, padding “0” means randomly choosing a certain number (w_{FA}^* or w_{FA}^s) of transmission slots in the odd-numbered slots within a period $T = L'$ and keeping silent in all the other slots.

As shown in Fig. 11, the MHUI sequence scheme still outperforms the baseline methods. For example, the average AAOI of the 8 users under the Optimal SA scheme (resp., Optimal FA scheme) is 203.1 (resp., 201.2), which is 17.3% (resp., 16.2%) higher than that under the MHUI sequence scheme, 173.2. Moreover, the duty factors of the Optimal SA and Optimal FA schemes are $f_{SA}^* = 0.0625$ and $f_{FA}^* = 0.0606$, respectively. They consume higher energy than the MHUI sequence scheme, which has duty factor $f_s = 0.024$. Under the Same SA scheme (resp., Same FA scheme) with duty factor same as f_s , the average AAOI of the 8 users is 220.8 (resp., 214.6), which is 27.5% (resp., 23.9%) higher than that under the MHUI sequence scheme.

D. Comparisons on Age Penalties

In the following, we will compare the age penalties under the MHUI sequence scheme and the baseline schemes. The simulation settings are the same as that in Subsection VIII-B. Fig. 12 shows the comparison result, in which the age penalty function is defined as $g(A_i(t)) = e^{aA_i(t)}$ with $a = 0.02$. We can see from Fig. 12 that the average age penalty under the sequence scheme is lower than those under the baseline schemes. For example, the average penalties of the 8 users under the Optimal SA scheme (resp., Optimal FA scheme) is 12.9 (resp., 12.3), which is 48.3% (resp., 41.4%) higher than that under the sequence scheme, 8.7. For the Same SA and Same FA schemes with the same duty factor as the sequence scheme, they lead to significantly higher age penalties. Specifically, the average penalties of the 8 users under the Same SA scheme (resp., Same FA scheme) is 18.2

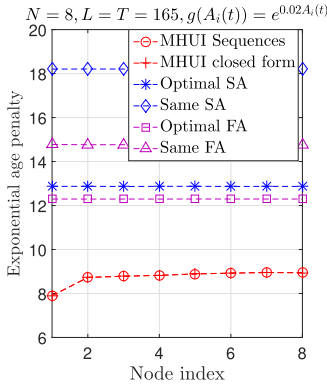


Fig. 12. Exponential age penalties under the MHUI sequence scheme and the baselines.

(resp., 14.8), which is 109.2% (resp., 70.1%) higher than that under the MHUI sequence scheme. In the meantime, we also plot the results of the derived closed-form expression for the MHUI sequence scheme. We can see that they are consistent with the simulation results.

E. AoI Performance With Unreliable Channel

In our previous discussions, we assume a perfect channel, which means that if two or more packets are transmitted at the same slot, they are considered lost. This simplification allows us to concentrate on the design of MAC protocol. Actual physical channels are more complex and include imperfection factors such as channel loss, noise, interference, and multipath. In this subsection, we show the AoI performance of our proposed sequence scheme and the baseline schemes when the channel is unreliable, to make the discussions more complete. The unreliable channel model we employed is commonly found in the literature [11], [18]: a collision-free packet will suffer an error and can be received successfully by AP with probability p_s . When $p_s = 1$, the channel is perfect.

As illustrated in Fig. 13, our proposed sequence scheme outperforms the baseline schemes, when p_s varies from 0.1 to 1. For example, when $p_s = 0.1$, which means that the channel quality is extremely poor, the average AAoI under the Optimal SA scheme (resp., Optimal FA scheme) is 284.1 (resp., 280.7), which is 14.6% (resp., 13.2%) higher than that under the sequence scheme, 248. Moreover, the two optimal ALOHA schemes also consume more energy than the sequence scheme due to their higher duty factors, as in the perfect channel case.

F. Comparisons in the Case of $T < L$

The simulation results shown previously are based on the condition that $T = L$. In the following, we will show the simulation results for the case that $T < L$. Fig. 14 below depicts the AAoI performance for the scenario in which T increases linearly from 25 to 125 with step size equal to 25, N is fixed at 13, and thus L is fixed at 325 according to the CRT construction. The case of $T = N = 13$ has also been shown in Fig. 14. The MHUI sequences are optimized under the guidance of Proposition 7. That is, for each T , we calculate the GCD α of T and L , and choose the optimal cyclic shift for

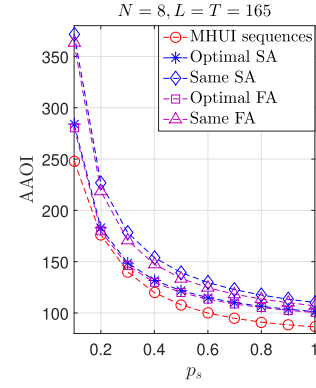


Fig. 13. AAoI performance under different channel qualities.

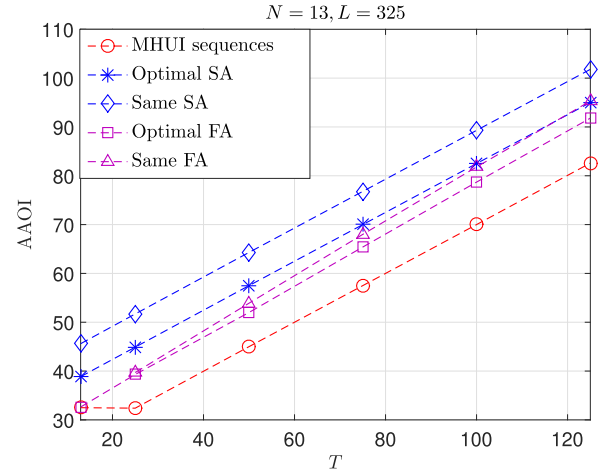


Fig. 14. AAoI performance under the MHUI sequence scheme and the baselines, when $T < L$.

each sequence from its α cyclic shifts. Note that in the case of $T = N$, for the framed ALOHA scheme, we only consider its optimal number of transmission attempts in each T , i.e., w_{FA}^* , since we cannot set a feasible value for w_{FA}^s to obtain the same duty factor as the sequence-based scheme.

As shown in Fig. 14, the MHUI sequence scheme achieves the lowest AAoI, as in the case of $T = L$. In the case of $T = N$, the AAoI performance under the MHUI sequences is close to that under the optimal framed ALOHA. However, in all other cases, the advantage of MHUI sequences over the other schemes is more obvious. For example, when $T = 25$, the AAoI values under the MHUI sequence scheme, the Optimal SA scheme and the Optimal FA scheme equal 32.4, 44.9 and 39.3, respectively. For all T in this scenario, the duty factors of the Optimal SA and Optimal FA schemes are $f_{SA}^* = f_{FA}^* = 0.0769$, which is nearly twice the duty factor $f_s = 0.04$ in sequence scheme. This indicates that the sequence scheme achieves higher energy efficiency.

Remark 3: We have conducted simulations to evaluate the relationship between the AoI performance and collision rate for $T \leq L$ (include $T = N$), where collision rate is defined as the ratio of failed transmission attempts due to collisions to all the transmission attempts. It reflects how frequent the collision happens. We have the following observations. At first, the collision rate is highly correlated with the duty factor.

Specifically, the more transmission attempts there are in a status generation period, the more collisions tend to occur. Second, when duty factor is fixed, the collision rate and AoI are positively correlated. We omit the simulation results due to page limit.

IX. CONCLUSION

In this paper, we consider sequence based MAC scheme for a system in which multiple users send periodically generated status information to a common AP over a collision channel. To the best of our knowledge, this is the first study that investigates AoI performance of sequence based MAC schemes. We have derived closed-form expressions for AAoI, APAoI and average age penalty under the proposed sequence scheme. Besides, we have revealed several critical properties to optimize the AoI performance. Slotted ALOHA and framed ALOHA are introduced and employed as baselines for performance comparison. Extensive simulations have been conducted to compare and evaluate our proposed sequence scheme against the two baseline schemes under various settings. Results show that the proposed sequence based scheme can attain enhanced AoI performance and also higher energy efficiency.

APPENDIX A PROOF OF THEOREM 2

Under the CRT construction, we have $w_1 = w_2 = \dots = w_N = N, L \approx 2N^2$. Thus,

$$P_{i,k} = \frac{1}{(2N)^{N-1}} \sum_{r=0}^{k-1} (-1)^{k-1-r} \binom{k-1}{r} \left((2N-k+r)^{N-1} - \sum_{n=0}^{k-2} \binom{N-1}{n} (2N-k)^{N-1-n} r^n \right).$$

For $r = 0, 1, \dots, k-1$,

$$\lim_{N \rightarrow \infty} \frac{(2N-k+r)^{N-1}}{(2N)^{N-1}} = \lim_{N \rightarrow \infty} \left(1 - \frac{k-r}{2N} \right)^{N-1} = e^{-\frac{k-r}{2}}.$$

For $n = 0, 1, \dots, k-2, r = 0, 1, \dots, k-1$,

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{\binom{N-1}{n} (2N-k)^{N-1-n} r^n}{(2N)^{N-1}} \\ = \lim_{N \rightarrow \infty} \frac{r^n}{2^n n!} \left(1 - \frac{k}{2N} \right)^{N-1} \\ = \frac{r^n}{2^n n!} e^{-\frac{k}{2}}. \end{aligned}$$

Then, we have

$$\begin{aligned} \lim_{N \rightarrow \infty} P_{i,k} &= \sum_{r=0}^{k-1} (-1)^{k-1-r} \binom{k-1}{r} \\ &\quad \left(e^{-\frac{k-r}{2}} - \sum_{n=0}^{k-2} \frac{r^n}{2^n n!} e^{-\frac{k}{2}} \right). \end{aligned}$$

Next, we will show that

$$\sum_{r=0}^{k-1} (-1)^{k-1-r} \binom{k-1}{r} \sum_{n=0}^{k-2} \frac{r^n}{2^n n!} e^{-\frac{k}{2}} = 0. \quad (24)$$

By simply exchanging the two summations in (24) we have

$$\begin{aligned} \sum_{r=0}^{k-1} (-1)^{k-1-r} \binom{k-1}{r} \sum_{n=0}^{k-2} \frac{r^n}{2^n n!} e^{-\frac{k}{2}} \\ = e^{-\frac{k}{2}} \sum_{n=0}^{k-2} \frac{1}{2^n n!} \left(\sum_{r=0}^{k-1} (-1)^{k-1-r} \binom{k-1}{r} r^n \right). \end{aligned}$$

For any function $f(x)$, we define an action Θ on $f(x)$ by taking the derivative with respect to x and then multiplying by x , i.e., $\Theta f(x) = x f'(x)$. Recall the Binomial Theorem: $(x-1)^{k-1} = \sum_{r=0}^{k-1} (-1)^{k-1-r} \binom{k-1}{r} x^r$. Then, $\Theta^n (x-1)^{k-1} = \sum_{r=0}^{k-1} (-1)^{k-1-r} \binom{k-1}{r} r^n x^r$. Therefore, $\sum_{r=0}^{k-1} (-1)^{k-1-r} \binom{k-1}{r} r^n = \Theta^n (x-1)^{k-1} |_{x=1}$. It is easy to see that, if $n \leq k-1$, then $\Theta^n (x-1)^{k-1} = (x-1) A(x)$ for some function $A(x)$. This implies that $\sum_{r=0}^{k-1} (-1)^{k-1-r} \binom{k-1}{r} r^n = 0$, whenever $n \leq k-1$. Finally, by letting $v = k-1-r$ and $\rho = e^{-\frac{1}{2}}$, we have $\lim_{N \rightarrow \infty} P_{i,k} = \sum_{v=0}^{k-1} (-1)^v \binom{k-1}{v} \rho^{v+1} = (1-\rho)^{k-1} \rho$. This completes the proof.

APPENDIX B PROOF OF LEMMA 7

Without loss of generality, we assume that $\delta_{i,1} = 0$. We use δ to represent $\delta_{i,2}$ for the brevity of notation below. Let $\delta = \ell\alpha$, where ℓ is an integer. We can also write $\mathbf{r}_i$ in the form of β/T status generation periods. That is, $\mathbf{r}_i = [\mathbf{t}_0 \dots \mathbf{t}_{\beta/T-1}]$, where $\mathbf{t}_k = [s_i(kT \bmod L) \dots s_i((kT + T - 1) \bmod L)]$ for $k \in \mathbb{Z}_{\beta/T}$. For $\mathbf{r}_i^\delta = [\underbrace{s_i^\delta \dots s_i^\delta}_{\beta/L}]$, it can also be written as

$\mathbf{r}_i^\delta = [\mathbf{t}_0^\delta \dots \mathbf{t}_{\beta/T-1}^\delta]$. We should prove that for any $y \in \mathbb{Z}_{\beta/T}$, we can find an $x \in \mathbb{Z}_{\beta/T}$ such that $\mathbf{t}_x = \mathbf{t}_y^\delta$. This is equivalent to showing that $\mathbf{r}_i(xT + u) = \mathbf{r}_i^\delta(yT + u)$, for $u \in \mathbb{Z}_T$. According to Bézout's identity, there must exist integers b, d such that $\alpha = bT + dL$. Therefore, $\mathbf{r}_i^\delta(yT + u) = s_i((yT + u + \delta) \bmod L) = s_i((yT + u + \ell(bT + dL)) \bmod L) = s_i(((y + b\ell)T + u) \bmod L)$. On the other hand, by letting $x = y + b\ell$, we have $\mathbf{r}_i(xT + u) = \mathbf{r}_i((y + b\ell)T + u) = s_i(((y + b\ell)T + u) \bmod L)$. Thus, $\mathbf{r}_i(xT + u) = \mathbf{r}_i^\delta(yT + u)$. This completes the proof.

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Fang Liu received the B.Eng. degree from Jilin University, China, in 2015, and the Ph.D. degree from the Department of Information Engineering, The Chinese University of Hong Kong, in 2020. She is currently a Post-Doctoral Fellow with the Department of Information Engineering, The Chinese University of Hong Kong. Her research interests include wireless communication networks, smart sensing, and the IoT applications.



Wing Shing Wong (Life Fellow, IEEE) received the combined bachelor's and master's degrees from Yale University and the M.S. and Ph.D. degrees from Harvard University. After working with the AT&T Bell Laboratories for ten years, he joined The Chinese University of Hong Kong in 1992, where he worked as the Chairman of the Department of Information Engineering from 1995 to 2003, the Dean of the Graduate School from 2005 to 2014, and the College Master of the S. H. Ho College from 2018 to 2022. He is currently an emeritus professor. He was a Science Advisor with the Innovation and Technology Commission of the HKSAR Government from 2003 to 2005. His research interests include communication to control. He is a fellow of the Hong Kong Academy of Engineering Sciences.



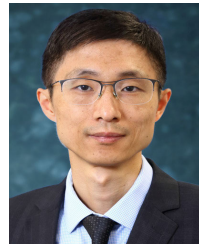
Yuan-Hsun Lo (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in applied mathematics from National Chiao Tung University in 2004, 2006, and 2010, respectively. He has been with Xiamen University since 2015. He joined National Pingtung University in 2019, where he is currently an Associate Professor with the Department of Applied Mathematics. His research interests include combinatorics, graph theory, and combinatorial designs and their applications.



Yijin Zhang (Senior Member, IEEE) received the B.S. degree from the Nanjing University of Posts and Telecommunications, China, in 2004, the M.S. degree from Southeast University, China, in 2007, and the Ph.D. degree from The Chinese University of Hong Kong in 2010, all in information engineering. He joined the Nanjing University of Science and Technology, China, in 2011. He is currently a Professor with the School of Electronic and Optical Engineering. His research interests include sequence design and resource allocation for communication networks.



Chung Shue Chen (Senior Member, IEEE) received the B.Eng., M.Phil., and Ph.D. degrees in information engineering from The Chinese University of Hong Kong (CUHK) in 1999, 2001, and 2005, respectively. He is currently a DMTS with the Nokia Bell Laboratories. Prior to joining the Bell Laboratories, he worked with INRIA, in the research group on Network Theory and Communications (TREC, INRIA-ENS). He was an Assistant Professor with CUHK. He was an ERCIM Fellow with the Norwegian University of Science and Technology (NTNU), Norway, and the National Centre for Mathematics and Computer Science (CWI), The Netherlands. He worked with CNRS, Lorraine, on Real-Time and Embedded Systems. His research interests include wireless communications and networking, optimization and algorithms, 5G/6G, the IoT, machine learning, and intelligent systems. He has served as a TPC in international conferences, including IEEE ICC, GLOBECOM, WCNC, PIMRC, VTC, CCNC, and WiOpt (the TPC Vice Chair). He is an Editor of the *Transactions on Emerging Telecommunications Technologies* (ETT) and an Associate Editor of the *Nature* (Springer) and *Telecommunication Systems*. He is a permanent member of the Laboratory of Information, Networking and Communication Sciences (LINCS), Paris. He was a recipient of the Sir Edward Youde Memorial Fellowship and the ERCIM Fellowship.



Guoliang Xing (Fellow, IEEE) received the B.S. and M.S. degrees from Xian Jiao Tong University, China, in 1998 and 2001, respectively, and the D.Sc. degree from Washington University in St. Louis in 2006. He is currently a Professor with the Department of Information Engineering, The Chinese University of Hong Kong. Previously, he was a Faculty Member with City University of Hong Kong and Michigan State University, USA. His research lies at the intersection between systems, embedded AI, data/information processing algorithms, and domain sciences, with a focus on interdisciplinary applications in health, environment, and energy. He is currently an Editor of *ACM Transactions on Sensor Networks* (TOSN) and has served as the Steering Committee Member, the General Chair, and the TPC Co-Chair for several international conferences, including IPSN: IEEE/ACMs Premier Conference on Information Processing in Sensor Systems and RTSS: IEEE's Premier Conference on Real-Time Systems.